

6 (a) (i) connection to 'top' of resistor labelled as positive B1 [1]

(ii) diode B and diode D B1 [1]

(b) (i) $V_P = 4.0\text{ V}$ C1
mean power = $V_P^2/2R$ C1
 $= 4^2 / (2 \times 2700)$
 $= 2.96 \times 10^{-3}\text{ W}$ A1 [3]

(ii) capacitor, correct symbol, connected in parallel with R B1 [1]

(c) graph: half-wave rectification M1
same period and same peak value A1 [2]

7 (a) wavelength associated with a particle M1